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# PhishStack: Evaluation of Stacked Generalization in Phishing URLs Detection

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## Abstract

Stacked Generalization has been assessed and evaluated in the field of Phishing URLs detection. This field has become egregious area of information security. Recently, different phishing URLs detection systems have already proposed by several researchers. But due to the lack of proper machine learning algorithm selection, the performance of those systems can be affected. A details investigation on individual machine learning classifiers on level 1 and final prediction from level 2 along with three real datasets have been presented on this paper. The performance has been evaluated by precision-recall curve, AUC-ROC curve, accuracy, misclassification rate and mean absolute error (MAE). The best AUC area obtained from Random Forest and Multi Layer Perceptron (MLP) individually. But stacked generalization provides higher accuracy of 97.44% with numeric feature set in binary classification and in multiclass feature set (dataset three), provides the performance with 97.86% of accuracy. Stacked generalization provides minimum error rate and MAE of 2.142857% with multiclass feature set which leads to a strong basement of developing an anti-phishing tools.

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**Keywords:** Phishing; Malicious URLs; Stacked Generalization; Anti-Phishing; Phishing Detection;

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## 1. Introduction

Phishing is a cyber scheme based on criminal activities which is acquiring attention of cyber attackers recently. Convincing users into exposing their login credentials, pin numbers, credit card information, personal data etc. refers to phishing. Through phishing, attackers can collect the information about the financial credentials to gain access and commits fraudulent activities [1, 2]. Uniform Resource Locator (URL) which is malicious plays significant role to perform phishing. The malicious URLs are being sent by cybercriminals to victims within a message using various

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communication mediums such as private chat message, emails, blogs, banners as well as on forums. The URLs which are sent look like a legitimate sources visually. According to an investigation, the amount of financial losses happened over 3 billion dollar annually by phishing attacks [3]. Thus, how phishing attacks occur and how can prevent have become a significant issues for security researchers. From some studies, three ways have been found as a proof of the occurrence of phishing. Duplication of a web interface of any trusted website is a way to perform phishing. Users will think that as a trusted site and will submit their sensitive information which refer to web based phishing. Secondly, email based phishing that is occurred by sending email and web based phishing also can involve within that email sent to victims. Injecting code which is malicious in a benign website and also by infecting with malware on the system used by the victims can be another way defined as malware based phishing [4, 5]. From those ways of phishing in both ways web based and email based phishing, the roles of URLs are noteworthy. That's why, phishing URLs detection has been considered here. There are multiple features already being identified to detect whether the URL is phishing or legitimate. There are various approaches have already been proposed by researchers. To identify the minimum features with maximum detection accuracy is the matter of concern now. The most relevant features have been selected based on features ranking scheme to detect the phishing URLs with machine learning algorithms. The core contributions of this paper can be stated as follows:

- A A multi-level approach (PhishStack) has been presented and evaluated the approach in the context of phishing URLs detection on a real publicly available dataset.
- B PhishStack can be performed with the combination of all types of classifiers such as machine learning, ensemble learning even also neural network based techniques as base classifiers.
- C PhishStack performed using stacked generalization which is a scheme for minimizing the generalization error rate of one or more generalizers.
- D PhishStack constructs based on a theme to minimize the error rate of generalizers.
- E Stacking concept has been applied to improve the accuracy of the phishing URLs detection. The performance of PhishStack has been evaluated to figure out the effectiveness of stacked generalization in Phishing URLs Detection.

The organization of the rest of the paper is constructed as: in section 2, Literature Review have been presented and discussed. The methodology of PhishStack and the dataset information which are used to experiment and evaluate has been described in section 3. In section 4, evaluation parameter and obtained results have been presented and discussed in details. Finally, Section 5 concludes the paper with a prospective future steps.

## 2. Literature Review

A real-time anti-phishing system has been proposed by Sahingoz, O. K., Buber, E., Demir, O., and Diri, B. which uses seven different classification algorithms along with different feature sets. They found that 97.98% accuracy rate for detection of phishing URLs by Random Forest algorithm with NLP based features which gives better performance [6]. PHISH-SAFE an anti-phishing system has proposed to detect phishing websites using SVM classifier with above 90% accuracy [7]. Gupta, S., and Singhal, A. claimed that comparatively random forest tree is a good technique to detect Phishing URLs in case of minimum execution time [8]. Parekh, S., Parikh, D., Kotak, S., and Sankhe, S. has been claimed that they have proposed a different methodology to detect phishing websites with Random Forest and found the accuracy level around 95% as highest [9]. Another anti-phishing technique based on a weighted URL tokens system by extracting identity keywords from a query webpage has been presented in one study. A search engine was invoked to pinpoint the target domain name using the identity keywords as search terms, and to determine the legitimacy of the query webpage. The model was proposed by Tan, C. L., and Chiew, K. L. proposed another anti-phishing technique by extracting keywords from a webpage which is followed by a weighted URL tokens based system [10]. An anti-phishing technique which is completely based on the URL (only six features) has been proposed and claimed as lightweight by Zouina, M., and Outtaj, B. [11]. They found 95.80% of accuracy using SVM. Mao, J., Tian, W., Li, P., Wei, T., and Liang, Z. have proposed based on visual similarity of web pages [12].

Features were extracted from the CSS (Cascading Style Sheet) of pages to measure the similarity of the suspicious pages and victim pages accurately. From similarity rating the effective feature sets were selected. They have evaluated their proposed approach by developing a prototype in Google Chrome browser.

Hybrid Ensemble Feature Selection (HEFS) based on a new feature selection framework with machine learning algorithm has been proposed by Chiew, K. L., Tan, C. L., Wong, K., Yong, K. S., and Tiong, W. K. [13]. A detection method based on the combination of Search, Heuristic Rule and Logistic Regression has been proposed by Ding, Y., Luktarhan, N., Li, K., and Slamu, W. [14]. Another phishing detection approach based on the combination of URL-based and Web page similarity based detection was proposed by Shekoka, N. M., Shah, C., Mahajan, M., and Rachh, S. [15]. LinkGuard algorithm has been used in their approach. A phishing attacks detection solution along with defending the zero-day phishing attacks has been proposed by Rao, R. S., and Ali, S. T. [16].

### 3. PhishStack: Presented Approach

The presented approach along with the assessment and obtained results will be discussed in this section.

#### 3.1. Dataset Used

There are three publicly available datasets from the UCI repository and Mendeley Data have been used for assessing this experiment. Dataset One (D1) has in total 11055 different URLs with 4,898 phishing and 6,157 legitimate URLs and for classification 30 features are considered [17]. Dataset two (D2) has 1353 URLs in numbered with only 10 features where 548 URLs are legitimate, 702 are phishing, and 103 suspicious [18]. Dataset three (D3) which is recently published has been used. D3 has total 10000 different URLs where 5000 URLs are phishing and rest 5000 URLs are the information of legitimate webpages. In this dataset total 48 features have been considered and which is downloaded within (January-May), 2015 and (May - June), 2017 [19]. The characteristics of attributes in D1 and D2 are integer types where D3 considered binary, categorical, continuous and discrete types of attributes.

#### 3.2. Used Machine Learning Algorithms

In this section, the machine learning algorithms which are considered for the experiment have been mentioned and discussed. Random Forest (RF), Decision Tree (DT), Multilayer Perceptron (MLP), Support Vector Machine (SVM), Stochastic Gradient Descent (SGD) and Gaussian Naive Bayes (GNB) are used as a base classifiers in level 1 of Stacked Generalization. 10-fold cross validation [20] has been used in level 1. To get final prediction from a meta estimator in level 2, XGBoost classifier is being used.

#### 3.3. The concepts of stacking

The evaluated approach PhishStack based on stacked generalization technique [21] has been depicted in Figure 1. In level 1, temporary predictions from the base classifiers along with cross validation are notated as  $P_{T1}$ ,  $P_{T2}$ , ...,  $P_{TN}$ . And by  $VP_{O1}$ ,  $VP_{O2}$ ,  $VP_{O3}$ , ...,  $VP_{ON}$  crisps predictions or output probability predictions are being expressed. For the computation during our model construction, the crisp predictions and the temporary predictions are being used in Level 2. The concepts [22, 27] evaluated in PhishStack are:

- A In Level 1, train and test set used to predict with base classifiers and after that on Level 2 the predictions obtained are being used as features.
  - B Any algorithms or techniques can be used in Level 1 or Level 2.
  - C To avoid overfitting for training set, N-fold cross-validation is used in presented model which can predict out-of-fold (OOF) of train part in every fold.
  - D 3-10 folds are usually being used
  - E Predict - Test Set:
- Level 1: After training, test set will be predicted and when it has done with all folds then need to apply or calculate the majority voting technique or mean (mode) of all temporary predictions  $P_{T1}$ ,  $P_{T2}$ , ...,  $P_{TN}$  from each fold.

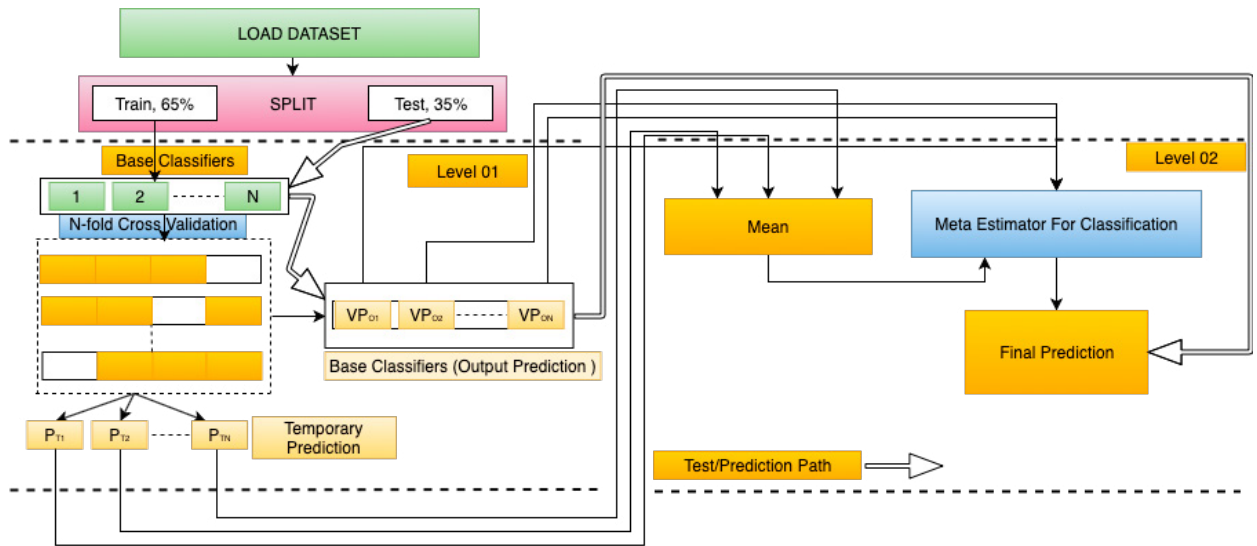


Fig. 1. PhishStack: Presented and Evaluated Approach

- Level 2: During n-fold cross validation, final prediction has not done for test set. When all folds cross validation complete then on Level 2 need to fit another additional classifier called meta-estimator on full train set and performs final predictions of test set. The approach takes much time rather than others because it performs additional fitting.

F Stacking concept has been implemented at figure 1 by applying 10-fold cross validation in Level 1. In Level 2, final prediction has been obtained.

To get more features for level 2 training and prediction, Level 1 is a repeated cycle.

## 4. Results and Discussion

### 4.1. Environmental Setup

The machine configuration in which the experiment has been conducted is Intel(R) Core(TM) i5-6500 CPU @ 3.20GHz processor, 64-bit PC with 16GB RAM. The operating system is Ubuntu 18.04.1 LTS (Bionic Beaver) and for simulation, python has been used as a well-established programming language along with scikit-learn [23], SciPy, pandas and numpy packages.

### 4.2. Evaluation Parameters

During this assessment, binary classification decision is being focused on whether the data is phishing or legitimate. A structure called confusion matrix can help to obtain the decision of the classifiers. Confusion matrix has four attributes: True negatives (TN): Correctly Identified phishing as phishing, False Negative (FN): Identified phishing as legitimate, False positives (FP): Identified legitimate as phishing, True positives (TP): Correctly Identified the legitimate as legitimate. Precision-Recall Curve, AUC-ROC Curve, Mean Absolute Error (MAE) and Accuracy has been used to evaluate the performance of classifiers. The evaluation metrics used in this assessment as tabulated in Table 1

Table 1. Evaluation Parameters for Assessment the classifiers

| Evaluation Parameter      | Description                                                                                                                                                                                                                                      | Formula                                                                                                                                                                          |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Accuracy                  | The ratio of correctly classified data to total data. [28]                                                                                                                                                                                       | Accuracy = (TP + TN) / (TP + TN + FP + FN)                                                                                                                                       |
| Precision - Recall Curve  | Precision - Recall Curve is plotted using the data pairs of recall and precision [24].                                                                                                                                                           | Positive Precision = TP/(TP + FP)<br>Negative Precision = TN / (TN+FN)<br>Positive Recall = TP/(TP + FN)<br>Negative Recall = TN/(TN+FP)                                         |
| AUC-ROC Curve             | AUC - ROC curve tells how much the model is capable of distinguishing between binary classes. The ROC curve is plotted with True Positive Rate - TPR against the False Positive Rate - FPR where TPR is on y-axis and FPR is on the x-axis [25]. | TRP (Positive Recall) = TP/(TP + FN)<br>FPR = 1- Specificity (Negative Recall)<br>= 1 - TN/(TN+FP) = FP/TN+FP                                                                    |
| Mean Absolute Error (MAE) | The average of all absolute errors [26].                                                                                                                                                                                                         | $MAE = \frac{1}{n} \sum_{i=1}^n  x_i - \hat{x} $<br>Where, n = the total number of errors,<br>Σ = summation symbol,<br> x <sub>i</sub> - x <sub>I</sub>   = The absolute errors. |
| Misclassification Rate    | Error Rate                                                                                                                                                                                                                                       | Misclassification Rate or Error Rate = 1 - Accuracy                                                                                                                              |

### 4.3. Performance Assessment and Result Analysis

There are three case studies have been experimented to evaluate the assessment. Three case studies such as Case Study #1, Case Study #2, Case Study #3 based on the evaluation of Stacked Generalization applied on dataset one (D1), dataset two (D2) and dataset three (D3) respectively.

#### 4.3.1. Case Study #1

Accuracy of base classifiers along with 10-Fold cross validation on D1

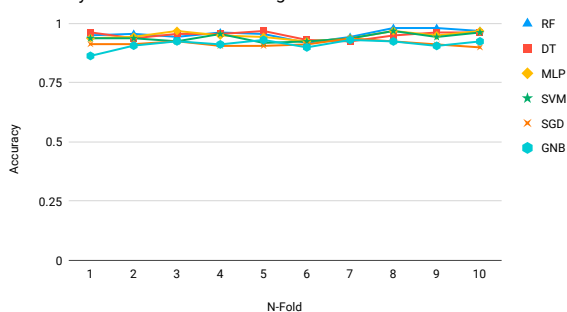


Fig. 2. Accuracy of base classifiers along with 10-Fold cross validation on D1

Accuracy and Misclassification Rate Obtained from Level 1 with D1

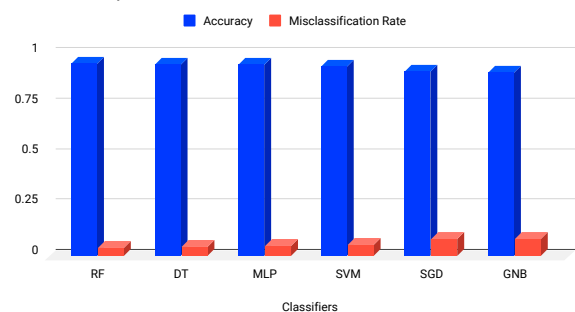


Fig. 3. Accuracy and Misclassification Rate Obtained from Level 1 with D1

Figure 2 represents the of base classifiers in according with the 10-fold validation. But the final accuracy and misclassification rate output from level 1 has been depicted at 3. The output from level 1 perform as a training features for next level. The precision-Recall curve and the ROC curve have been presented by figure 4 and 5. It's been found that in level 1, maximum 95.61% of accuracy attained from random forest individually with minimum error rate. In case of precision-recall curve and the AUC-ROC curve, stacked generalization provides low performance where RF and MLP do better individually. But the final prediction accuracy from stacked generalization has been found 97.44% which is higher than individual classifiers.

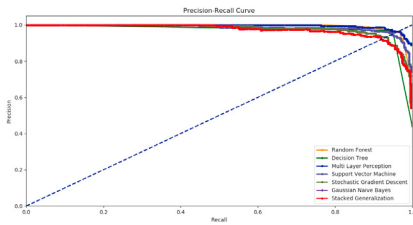


Fig. 4. Precision-Recall Curve of D1

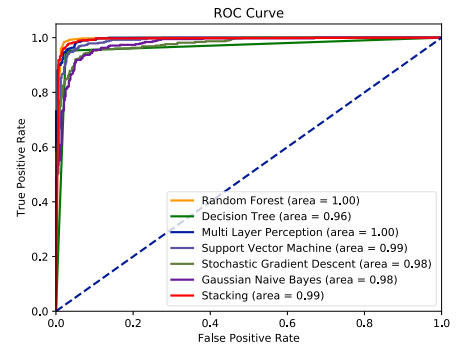


Fig. 5. ROC Curve of D1

#### 4.3.2. Case Study #2

The accuracy comparison of base classifiers along with D2 obtained from each fold of cross validation has been depicted at Figure 6. The output final accuracy and error rate from level 1 has been visualized through Figure 7. Level 2 has been trained by using the output predictions from level 1 which are known as stacking features. Figure 8 and 9 depict the precision-Recall curve and the ROC curve respectively. In level 1, maximum 96.72% accuracy has been found from random forest individually and also minimum error rate. And the final prediction 96.77% of accuracy from stacked generalization has been found which is little bit higher than individuals. Stacked generalization has little low performance in terms of precision-recall curve but in case of the ROC curve, stacked generalization, RF, MLP and SVM provides same AUC of 99%.

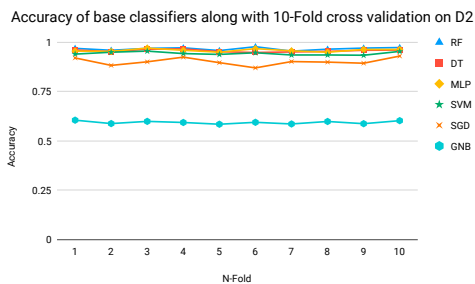


Fig. 6. Accuracy of base classifiers along with 10-Fold cross validation on D2

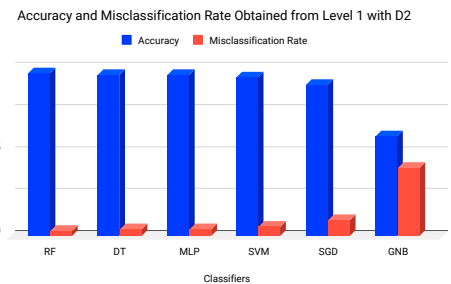


Fig. 7. Accuracy and Misclassification Rate Obtained from Level 1 with D2

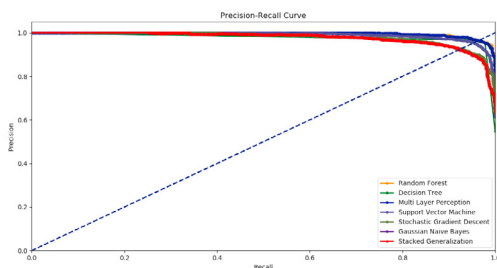


Fig. 8. Precision-Recall Curve of D2

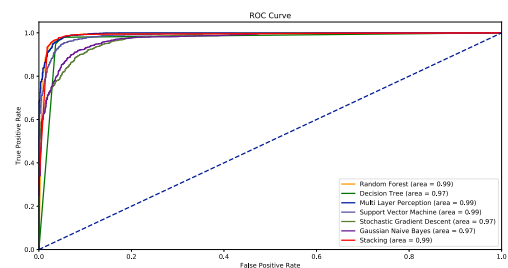


Fig. 9. ROC Curve of D2

### 4.3.3. Case Study #3

The comparison of the accuracy obtained from each fold of base classifiers along with D3 has been depicted at Figure 10. Figure 11 represents the output final accuracy and error rate from level 1 has been visualized. Level 2 has been trained by using stacking features which are the output predictions from level 1. The precision-Recall curve and the ROC curve has been represented by figure 12 and 13 respectively. In level 1, minimum error rate with maximum accuracy of 97.63% has been found from random forest individually. But stacked generalization provides 97.86% of accuracy. Stacked generalization provides better performance in terms of precision-recall curve as well as in case of the AUC-ROC curve with 99% of AUC area.

Accuracy of base classifiers along with 10-Fold cross validation on D3

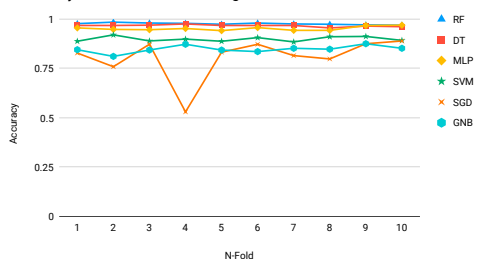


Fig. 10. Accuracy of base classifiers along with 10-Fold cross validation on D3

Accuracy and Misclassification Rate Obtained from Level 1 with D3

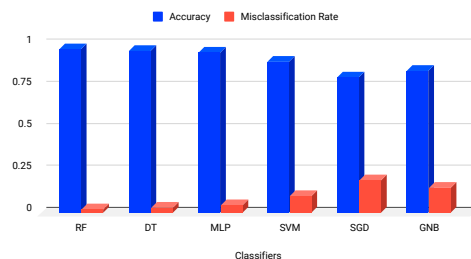


Fig. 11. Accuracy and Misclassification Rate Obtained from Level 1 with D3

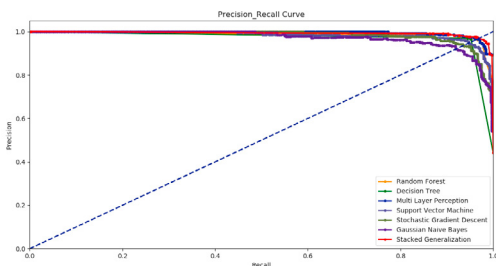


Fig. 12. Precision-Recall Curve of D3

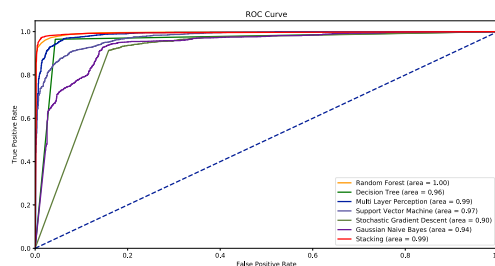


Fig. 13. ROC Curve of D3

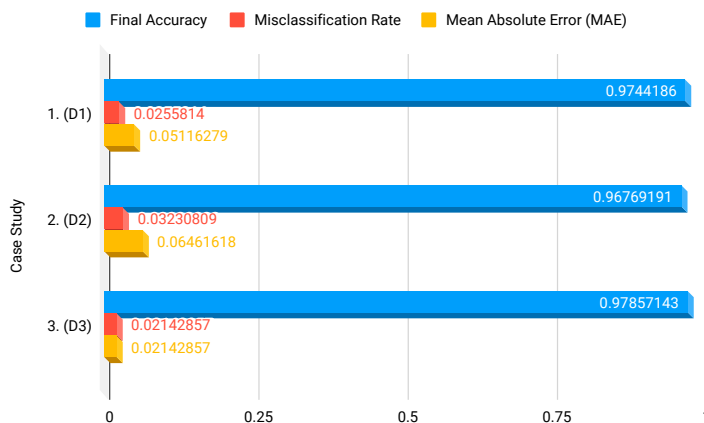


Fig. 14. Final Accuracy, Misclassification Rate and Mean Absolute Error (MAE)

Figure 14 depicts the final accuracy, error rate and MAE of all of three cases. Stacked Generalization provides better accuracy with less error rate and MAE for all three cases. It has been also observed that D1 with minimum 10 attributes, stacked generalization provides around 97.44% accuracy, 2.55814% error rate, 5.116279% MAE. On the other hand, with the most recent dataset D3 where in total 48 attributes were considered stacked generalization provides highest accuracy of 97.86% with the same error rate and MAE of 2.142857%.

## 5. Conclusion

This assessment will lead the effectiveness of stacked generalization during to develop an anti-phishing tools. The basement of developing strong phishing detection tools can be lead by this investigation. It has been noticed that Random Forest and Multi Layer Perceptron (MLP) provide better precision and recall but stacked generalization performs better in terms of overall accuracy. For stacked generalization based binary classification detection tools, D1 with 10 features provides more accurate decision rather than D2 which has 30 features. Stacked generalization provides 97.44% and 97.86% of accuracy with dataset one and three accordingly. But stacked generalization gets better learning base from the latest dataset with more features in total 48 features. Another findings is stacked generalization performs better with multiclass dataset D3 rather than numeric type dataset D1 and D2. In recapitulate, the performance of stacked generalization provides a strong basement to the development of anti-phishing tools. In future, the real life phishing detection tools will be developed for continuing this study.

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